O-G-491C
14 November 1983
SUPERSEDING
Fed. Spec. O-G-491b
January 30, 1967

FEDERAL SPECIFICATION

GLYCEROL, TECHNICAL (HIGH GRAVITY)

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one grade of glycerol also known as high-gravity glycerin.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
TT-E-485	-	Enamel, Semi-Gloss, Rust-Inhibiting
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging Of
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Federal Standards:

Fed. Std. No. 123	-	Marking for Shipment (Civil Agencies)
Fed. Std. No. 595	-	Colors

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1168 - Ammunition Lot Numbering

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM Standards:

D 1193 - Reagent Water
D 1258 - Sampling and Testing High-Gravity Glycerin
D 3951 - Commercial Packaging

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(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Physical and chemical characteristics. Glycerol shall conform to the physical and chemical characteristics of table I when tested as specified herein.

TABLE I. Physical and chemical characteristics

Characteristic	Requirement	Test paragraph
Appearance	Clear and free of suspended matter	4.2.4.1
Odor	Slight, characteristic	4.2.4.1
Color, Pt-Co max	20	4.2.4.1
Specific gravity at 25°/25°C	1.2587	4.2.4.1
Acidity or alkalinity	To pass test	4.2.4.2
Ash, percent by weight, max	0.10	4.2.4.3
Chlorides calculated as Cl, percent by weight, max	0.01	4.2.4.4
Saponification equivalent, percent by weight, max	0.05	4.2.4.5
Silver nitrate precipitation	No precipitate formed in 10 minutes	4.2.4.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the glyerol produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). When specified (see 6.2), lot numbering shall be in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For glycerol test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 2 liters shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for glycerol test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard liners, partitions, or pads missing or not as specified (when required)
(f) Container coating missing or not as specified (when required)
(g) Marking incorrect, missing or illegible
(h) Unitization not as specified

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4.2.3.2 For glycerol test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 Glycerol tests. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance, odor, color, and specific gravity. Determine in accordance with the applicable test procedure in ASTM D 1258.

4.2.4.2 Acidity or alkalinity. Measure 50 mL of the specimen in a 50-mL graduated cylinder and pour into a 200-mL Erlenmeyer flask. Rinse the graduated cylinder with 100 mL of carbon dioxide-free water that has been rendered exactly neutral to phenolphthalein indicator and pour the rinsings into the flask containing the specimen. Titrate the glycerin solution to neutrality with 0.5N hydrochloric acid (HCl) or 0.5N sodium hydroxide solution (NaOH), as required, using phenolphthalein indicator. Multiply the number of milliliters required by the normality of the solution used and report the product as milliliters of normal solution required for 50 mL of the specimen. To pass test, 50 mL of specimen shall require not more than 0.30 mL of 1N HCl or of 1N NaOH for neutralization.

4.2.4.3 Ash. Determine the ash content of the specimen in accordance with the sulfate ash procedure in ASTM D 1258. Retain the residue for the chloride determination (see 4.2.4.4).

4.2.4.4 Chlorides, calculated as Cl. Add about 25 mL of hot water to the ash residue obtained in 4.2.4.3 and rub with the flattened end of a stirring rod. Add 1 mL of a solution made by dissolving 10 g of potassium chromate in 100 mL of water and titrate with 0.01N silver nitrate solution to the first permanent reddish tint. Run a blank and calculate the percent by weight total chlorides present as follows:

$$\text{Percent total chlorides} = \frac{3.546\ VN}{W}$$

where: V = Volume of standard silver nitrate solution (corrected for blank), in milliliters,
N = Normality of silver nitrate solution, and
W = Weight of specimen used in 4.2.4.2, in grams.

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4.2.4.5 Saponification equivalent. Transfer a 25-g portion of the specimen to a 250-mL Erlenmeyer flask. Add 100 mL of freshly boiled water. Pipet 10 mL of 0.33N sodium hydroxide solution into the flask. Attach the flask to a condenser and reflux the contents gently for 1 hour. Remove the flask and contents from the source of heat, allow the contents to cool to room temperature, and titrate the excess sodium hydroxide with 0.1N hydrochloric acid, using phenolphthalein as indicator. Run a blank. Calculate the saponification equivalent as percent by weight sodium oxide as follows:

$$\text{Saponification equivalent (as percent } Na_2O) = \frac{3.10\ (A - B)\ N}{W}$$

where: A = Volume of hydrochloric acid used for blank, in milliliters,
B = Volume of hydrochloric acid used for specimen, in milliliters,
N = Normality of hydrochloric acid solution, and
W = Weight of specimen used, in grams.

4.2.4.6 Silver nitrate precipitation. Pipet a 10-mL portion of the specimen to a 25-mL glass-stoppered cylinder and add exactly 10 mL of a 10.0 percent solution of silver nitrate. Mix the solution thoroughly by repeated inversion of the cylinder. (Note: The mixing operation should not be so vigorous as to produce a froth. Both the addition of the silver nitrate solution and the mixing operation should be done in subdued light.) Allow the solution to stand for 10 minutes and then examine the contents of the cylinder for the formation of a flocculent precipitate. If a flocculent precipitate (not to be confused with a finely divided precipitate of silver) is formed, the glycerol shall be considered as having failed this test. (If silver oxide precipitate forms, specimen should be neutralized to pH 7.0 prior to testing.)

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes.

(a) Upright
(b) Upside down
(c) On one side (or one quadrant)
(d) On one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakage.

5. PACKAGING

5.1 Unit packing. Glycerol shall be unit packed level A, B or industrial, as specified (see 6.2), in accordance with the general requirements of PPP-C-2020.

5.1.1 Level A. Glycerol shall be unit packed level A in a 1- or 55-gallon (gal) quantity as specified (see 6.2).

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5.1.1.1 One-gal quantity. A quantity of 128 (+2 or -0) fluid ounces of glycerol shall be unit packed in a nominal 1-gal capacity can conforming to type V, class 4, oblong, tin-plate of PPP-C-96. The can shall be formed from minimum number 100 electrolytic tin-plate, shall be furnished with a bridge type handle, and shall have exterior nonmetallic coating, plan B with side seam striped. The closure shall be in addition to an inner seal and shall be closed to a torque within the range specified by the can supplier. There shall be no evidence of leakage of contents when tested as specified in 4.2.5.

5.1.1.2 Fifty-five-gal quantity. A quantity of 55 (+1/2 or -0) gal of glycerol shall be unit packed level A in nominal 55-gal steel drum conforming to Rule 40 of the Uniform Freight Classification except that the entire interior surface of the drum shall be furnished with two coats of baked-on epoxy-phenolic enamel. The exterior shall be finished as specified for the type II drum of PPP-D-729, using olive drab enamel color No. 24087 of Fed. Std. No. 595, conforming to TT-E-485 or equal. Also, each plug on the drum shall be furnished with a cap-seal. The plugs shall be closed to a torque within the range specified by the drum manufacturer. There shall be no evidence of leakage of contents from the drum when tested as specified in 4.2.5.

5.1.2 Level B. Glycerol shall be packed level B in a 1- or 55-gal quantity as specified (see 6.2).

5.1.2.1 One-gal quantity. The one-gal quantity of glycerol shall be unit packed level B in the same manner as for level A protection in 5.1.1.1, except that no exterior nonmetallic coating shall be required.

5.1.2.2 Fifty-five-gal quantity. The 55-gal quantity of glycerol shall be unit packed level B in the same manner as for level A protection in 5.1.1.2, except that commercially acceptable exterior coating of the drum shall be acceptable.

5.1.3 Industrial. A specified quantity of glycerol (see 6.2) shall be unit packed in accordance with ASTM D3951.

5.2 Packing. Glycerol shall be packed level A, B, or industrial as specified (see 6.2).

5.2.1 Level A. Glycerol shall be packed level A as four 1-gal cans or a 55-gal drum.

5.2.1.1 One-gal quantity. Four 1-gal cans of glycerol, unit packed as specified in 5.1, shall be packed level A in a close-fitting wirebound box. The box shall conform to class 3, style optional of PPP-B-585, with wood parts preserved as specified for the grade A box of PPP-B-621. All inside faces of the box shall be lined with fiberboard, and each can shall be placed in a close-fitting full-can-height cell formed from half-slotted partitions. Motion of contents shall be prevented by inserting fiberboard pads where needed. Liners, partitions, and pads shall be formed from fiberboard conforming to grade V3c of PPP-F-320. The box shall be closed as specified in PPP-B-585.

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5.2.1.2 Fifty-five-gal quantity. The 55-gal quantity of glycerol, unit packed as specified in 5.1, shall require no further protection for shipment.

5.2.2 Level B. Glycerol shall be packed level B as four 1-gal cans or a 55-gal drum.

5.2.2.1 One-gal quantity. The one-gal quantity unit packs of glycerol shall be packed level B in the same manner as level A in 5.2.1.1, except that the box shall conform to weather-resistant grade V3c of PPP-B-636, and the box shall be reinforced after closure, in accordance with the appendix to PPP-B-636.

5.2.2.2 Fifty-five-gal quantity. The 55-gal quantity of glycerol, packed level B as specified in 5.1.1.2, shall require no further protection for shipment.

5.2.3 Industrial. Glycerol shall be packed industrially in accordance with ASTM D3951.

5.2.4 Unitization. Uniform quantities of level A and B packs of the 1-gal quantity of glycerol shall be palletized in accordance with load type I of MIL-STD-147 using the soft wood pallet conforming to type IV of NN-P-71. Uniform quantities of industrial packs of glycerol shall be unitized in a manner to assure protection of the glycerol from supply source to initial destination, acceptance by common carrier, and conformance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or the rules applicable to any other intended mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Level A and B unit packs, packs and unitized loads shall be marked in accordance with MIL-STD-129. Industrial unit packs and packs shall be marked in accordance with ASTM D3951. Industrial unitized loads shall be marked to show proper identification and to assure timely delivery from supply source to initial destination. In addition, each container shall be marked to show lot or batch number and date of manufacture of the glycerol.

6. NOTES

6.1 Intended use. The glycerol covered by this specification is intended for use in the manufacture of explosives and in other industrial applications for which high-gravity glycerol is required.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Level of unit packing and packing required (see 5.1 and 5.2),

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(c) Quantity to be unit packed level A, B, or industrially (see 5.1.1, 5.1.2, and 5.1.3), and
(d) If lot numbering in accordance with MIL-STD-1168 is required (see 4.2.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.5 Caution. Glycerol is a very hygroscopic material and precautions should be taken to prevent it from absorbing moisture from the atmosphere. If possible, sampling should be made in clear, dry weather.

MILITARY INTERESTS:

Custodians:

Army - EA
Navy - OS
Air Force - 68

Review activities:

Army - MD, MI
DLA - GS

Project No. 6810-0818

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
O-G-491C	GLYCEROL, TECHNICAL (HIGH GRAVITY)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* -- Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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